A balance-consistent restart for the moving-cavity mesh change

Deriving the conditions newly-opened and newly-iced cells must satisfy for settle-free continuation

AWI-ESM3 $\leftrightarrow$ PISM moving cavity — design note

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Abstract

On every moving-cavity leg the FESOM restart is remapped onto a sub-mesh whose ice front has moved. Empirically the leg then blows up within days: *fast* as a vertical-CFL runaway at $\Delta t = 1200\,\mathrm{s}$, or *slow* as a barotropic $\eta_n \to$NaN at $\Delta t = 60\,\mathrm{s}$. We have falsified every *value-only* restart fix (removing static density inversions; nearest-neighbour velocity; running without a settle). This note shows that the two blow-ups are the two faces of a single **geostrophic adjustment**, that the balanced state a stable restart must already be in is **fully determined** by the density and sea-surface height together with the surrounding (already-spun-up) water masses, and that we can therefore **compute** the required velocity (thermal wind + barotropic reference) rather than guess it. The corollary is that the remap must also not over-sharpen the opened-column fronts. The plan is to build both into the F90 remap and verify the balance metrics *before* the run, giving settle-free continuation.

1 Status: what is falsified

The instability is a coastal, mid-column ($\sim$95 m) mode that grows over days from the changed margin. Ruled out, each verified at the data level and each leaving the blow-up essentially unchanged:

- **Static density inversions** — a coherent-column T/S refill drove the worst stability over all retreated cells from -8.8 to $-0.5\,\mathrm{kg\,m^{-3}}$ (11 severe $\to$ 0); the leg still blew (`mstep`$\approx$164, CFLz$\approx$8.5).
- **Nearest-neighbour velocity** (Finn Heukamp / SO-ASE handling) — filling changed elements with the nearest old element's current (no cell left at zero) lowered peak CFLz 8.5 $\to$ 5.7 but *still* blew at `mstep`$\approx$164, now with a barotropic $\eta_n \to$NaN as well.
- **No settle / timestep** — $\Delta t = 60\,\mathrm{s}$ removed the fast CFLz mode entirely (0 warnings, $\sim$10 days) but a slow barotropic η_n mode grew and NaN'd.

A value-only restart cannot be made *balanced*: the imbalance is lateral (fronts) + missing momentum, not a vertical inversion. We now derive what "balanced" means quantitatively.

2 The balance a stable restart must satisfy

Started from a density field, the primitive equations relax to geostrophic–hydrostatic balance; "no adjustment shock" is equivalent to the restart already satisfying, at every wet point:

$$\text{(1) static stability:} \quad N^2 = -\tfrac{g}{\rho_0}\,\partial_z\rho \;\geq\; 0, \tag{1}$$

$$\text{(2) hydrostatic:} \quad p(z) = \rho_0 g\,\eta \;+\; \underbrace{\rho_i g\,h_i}_{\text{ice-shelf load}} \;+\; \int_z^0 \rho\,g\,dz', \tag{2}$$

$$\text{(3) geostrophy:} \quad f\,\hat{\mathbf{z}}\times\mathbf{u} = -\tfrac{1}{\rho_0}\nabla_h p, \tag{3}$$

$$\text{(4) thermal wind:} \quad \partial_z\mathbf{u} = -\tfrac{g}{f\rho_0}\,\hat{\mathbf{z}}\times\nabla_h\rho, \tag{4}$$

$$\text{(5) barotropic reference:} \quad \mathbf{u}(z{=}0) = \tfrac{g}{f}\,\hat{\mathbf{z}}\times\nabla_h\eta, \tag{5}$$

$$\text{(6) non-divergence:} \quad \nabla_h\cdot\mathbf{u}\approx 0. \tag{6}$$

(1) and (6) are the two we *watch*: (1) keeps convection off, and (6) — automatically satisfied by geostrophic flow to $O(\text{Rossby})$ — keeps the diagnosed w (hence CFLz) *and* $\partial_t\eta$ (hence the barotropic NaN) small. So a state that satisfies (3)–(5) kills *both* the fast and the slow blow-up at once.

The velocity is computable, not guessable. Integrating (4) from the surface and adding the barotropic reference (5),

$$\boxed{\mathbf{u}(z) \;=\; \underbrace{\tfrac{g}{f}\,\hat{\mathbf{z}}\times\nabla_h\eta}_{\text{barotropic, from }\eta} \;+\; \underbrace{\tfrac{g}{f\rho_0}\int_z^0 \hat{\mathbf{z}}\times\nabla_h\rho\;dz'}_{\text{thermal wind, from }\rho(T,S)}} \tag{7}$$

with the vertical integration constant / reference fixed by the **surrounding unchanged cells**, which are already in the model's balanced state. Set T, S, η from the neighbours; then $\mathbf{u}$ is *determined* — it must not be zeroed (v4) nor NN-copied (v5).

3 Quantitative diagnosis: the missing jet

At the Weddell seed the fill leaves a front of $\Delta S\approx 3\,\text{psu}$ over $\Delta x\approx 25\,\text{km}$, so $\Delta\rho\approx\beta_S\Delta S\approx 2.4\,\text{kg}\,\text{m}^{-3}$ and $\nabla_h\rho\approx 1.0\times10^{-4}\,\text{kg}\,\text{m}^{-4}$. With $f(-66^\circ)\approx -1.33\times10^{-4}\,\text{s}^{-1}$, (4) gives

$$\partial_z u = \tfrac{g}{f\rho_0}\,\nabla_h\rho \approx 7\times10^{-3}\ \text{s}^{-1} \;\Longrightarrow\; \Delta u\approx 0.7\ \text{m}\,\text{s}^{-1}\ \text{over 100 m}. \tag{8}$$

The balanced front carries a $\sim$0.7 m s^{-1} geostrophic jet. We initialised it at 0 (v4) or $\sim$0.06 m s^{-1} (v5): an $O(0.7\,\text{m s}^{-1})$ momentum deficit — precisely the violent adjustment observed. NN cannot supply it because the required current is specific to *this* density gradient, not the neighbour's background flow.

4 Conditions per case

Added points (ice retreated / newly wet). At each newly-wet level:

1. T, S: a statically stable ($N^2\geq 0$) blend of the surrounding water masses at that depth — no grid-scale front sharper than the physical one.

2. $\mathbf{u}$: the thermal-wind profile of Eq. (**??**) implied by that T/S, referenced to the neighbouring balanced current.
3. η: smooth to the neighbours (hydrostatically consistent).
4. `hnode`: ALE-consistent (sums to $H + \eta$), so no spurious $\partial_t h \to w$.

Points next to removed points (ice advanced / newly iced). The new ice adds a surface load $\rho_i g h_i$, so η at the iced cell drops hydrostatically by $\approx (\rho_i/\rho_0)h_i$. The *adjacent* open cells must see that pressure step as a *balanced* geostrophic current: their $\mathbf{u}$ must include the geostrophic response (5) to the new ice-loading slope. Computable from the PISM draft h_i and hydrostaty.

5 Corollary: do not over-sharpen the front

A $0.7\,\mathrm{m\,s^{-1}}$ jet is high for an Antarctic cavity front, which flags that the current fill **over-sharpens** it: the shelf-base "hold upward" plants $S \approx 31$ through the whole newly-opened column, where physically only a thin near-base melt layer is that fresh. The two levers are therefore linked:

gentler T/S front $\Rightarrow$ smaller $\nabla_h \rho$ $\Rightarrow$ smaller required jet; set that jet by thermal wind $\Rightarrow$ the remainder

Do both and conditions (1)–(6) hold at $t = 0$ by construction $\Rightarrow$ no adjustment $\Rightarrow$ **settle-free continuation**, i.e. the full year to the next PISM call at 1200 s.

6 Implementation plan (F90 remap, `mo_remap_fields.F90`)

1. **Soften the opened-column T/S.** Replace the shelf-base "hold upward" with a distance/depth-weighted blend toward the surrounding open-ocean profile (keep real cavity water near the base; relax the mid/upper column), preserving $N^2 \geq 0$.
2. **Density gradient.** On the new sub-mesh compute $\nabla_h \rho$ per level from the (softened) T, S via the finite-volume gradient FESOM already uses (nabla of a nodal field on the triangulation); use a linearised EOS locally (sufficient for the gradient).
3. **Thermal-wind integral.** Integrate Eq. (**??**) downward from the surface on each changed *element*, with the integration reference set to the nearest unchanged element's velocity (the SO-ASE NN, now used only as the *barotropic reference*, not the whole field).
4. **Barotropic part.** Add $\frac{g}{f}\hat{\mathbf{z}} \times \nabla_h \eta$ from the (donor-filled, ice-loaded) η; near-coast f is safely large (Antarctic), no equatorial issue.
5. **Verify *before* running.** Report, over the changed cells: $\min N^2$, $\max |\nabla_h \cdot \mathbf{u}|$ (target $\ll$ the CFL bound), and the thermal-wind residual $|\partial_z \mathbf{u} + \frac{g}{f\rho_0}\hat{\mathbf{z}} \times \nabla_h \rho|$. These are cheap, and a green check here predicts a stable leg.
6. **Re-test** the movcav8 chunk-3 leg at 1200 s, no settle.

7 Caveats / open points

- **Discrete divergence.** The mesh-discrete geostrophic velocity has a small residual $\nabla_h \cdot \mathbf{u}$; it should sit below the CFL bound but must be checked (step 5).
- **Barotropic–baroclinic consistency.** η and the density's steric contribution must be mutually consistent; if the donor η and the thermal-wind interior disagree at the reference level, prefer the neighbour-matched value and let η follow.

- **Ageostrophic part.** Tidal/Ekman components are not in (3)–(5); at $t = 0$ for a spun-up interior geostrophy dominates, and matching the neighbour current as the reference carries the ageostrophic background across.
- **Still-sub-ice retreat cells** (e.g. ulev $14 \to 2$, the actual seed) are inside the cavity, not open ocean; their $\mathbf{u}$ reference is the neighbouring *cavity* flow, and their T/S blend is toward cavity, not open-ocean, water.
- **Source term.** If the per-leg front motion is itself implausibly large (deep-draft open$\leftrightarrow$cavity transitions in one 10-yr PISM step), gentling the `cavity_nlvls` regeneration reduces $\nabla_h \rho$ at the root and makes every step above easier.